LIFE EXPECTANCY & GDP ANALYSIS

Exploring Economic Growth and Population Health Across Six Countries (2000 – 2015)

A Python-Based Exploratory Data Analysis Case Study

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Project summary

This case study examines how Gross Domestic Product (GDP) and life expectancy changed between 2000 and 2015 in Chile, China, Germany, Mexico, the United States of America, and Zimbabwe. Using Python, pandas, Matplotlib, and Seaborn, the analysis explores country-level trends, compares changes over time, and investigates the association between economic output and life expectancy.

The objective is to demonstrate how a structured data-analysis workflow can turn a small dataset into interpretable visualizations and evidence-based observations.

EXECUTIVE SUMMARY

Objective

Economic performance and population health are important indicators of national development. However, changes in economic output do not necessarily correspond directly to changes in life expectancy.

This project investigates three questions:

1. How did life expectancy change across the six countries between 2000 and 2015?
2. How did GDP change over the same period?
3. Did countries with greater GDP growth also experience greater improvements in life expectancy?

Key findings

Finding 1 - **Life expectancy trends**: The analysis examines changes in life expectancy over time across all six countries.

Result: Zimbabwe displays a distinct pattern, with a slight decline in the early 2000s followed by a sustained and much steeper increase in later years.

Finding 2 — **Economic growth**: The GDP analysis examines changes in economic output and differences in the trajectories of the six countries.

Result: China's GDP showed consistent and substantial growth throughout the study period, exhibiting the steepest upward trend among the six countries.

Zimbabwe's GDP declined during the early 2000s and reached its lowest point around 2008. After that, GDP increased steadily through 2015, indicating a period of economic recovery.

Finding 3 - **GDP growth and life expectancy improvement**: A country-level comparison examines percentage GDP change alongside the absolute change in life expectancy.

Result: Zimbabwe achieved the largest improvement in life expectancy (+14.7 years), despite GDP growth (143.7%) being much lower than China's. China recorded the largest GDP growth (813.4%), yet its life expectancy increased by 4.4 years, which was smaller than Zimbabwe's improvement.

Why these findings matter

The comparison demonstrates the value of examining multiple indicators when assessing development. GDP describes economic output, while life expectancy provides an indicator of population health. Their relationship may reflect a range of economic, social, healthcare, demographic, and policy factors.

The analysis is exploratory. It identifies patterns in the available data but does not establish that changes in GDP caused changes in life expectancy.

DATASET AND METHODOLOGY

Dataset overview

The project uses a country-level dataset obtained from Codecademy.

Attribute	Description
Observations	96 rows
Variables	4 columns
Countries	6
Time period	2000 – 2015
Observations per country	16
Missing values	0
Duplicate rows	0
Duplicate country-year combinations	0

Countries included

Chile, China, Germany, Mexico, the United States, and Zimbabwe.

Variables

- **Country**: Identifies the country associated with each observation.
- **Year**: Identifies the year of the observation.
- **Life expectancy at birth (years):** Represents life expectancy at birth, measured in years.
- **GDP**: Represents gross domestic product (GDP) as defined in the source dataset.
  Its units and whether it is nominal or inflation-adjusted should be confirmed from the original dataset documentation before making precise economic comparisons.

Data Quality Assessment

The initial inspection examined the dataset's dimensions, column names, data types, missing values, duplicate rows, country coverage, and year range.

The dataset contains 96 observations, corresponding to six countries observed over 16 years. The inspection identified no missing values, duplicate rows, or duplicate country-year combinations.

These checks provide confidence that the dataset contains complete and consistent country-year coverage. They do not, by themselves, establish the accuracy or comparability of the original measurements.

Analytical methodology

The analysis follows a structured exploratory workflow:

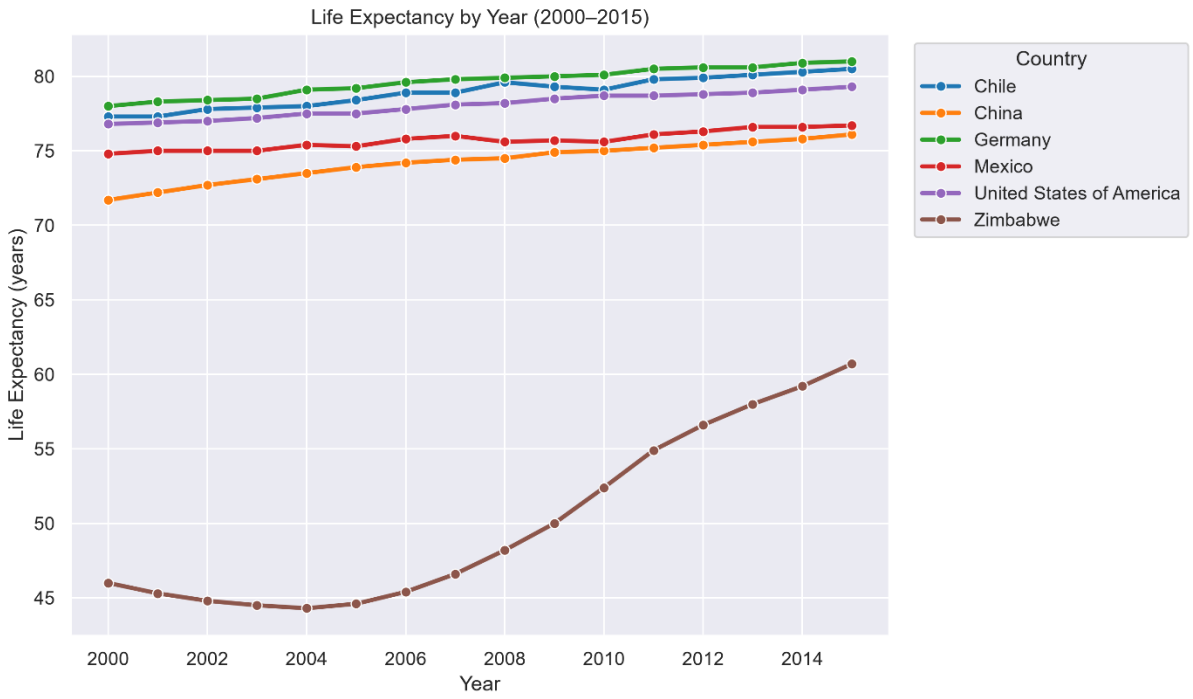
1. Load and inspect the dataset using pandas.
2. Assess data quality and prepare the data for analysis.
3. Compare life expectancy trends across countries.
4. Examine GDP trends over time, using a logarithmic scale where appropriate.
5. Compare life expectancy improvements and GDP changes between 2000 and 2015.
6. Explore the relationship between GDP and life expectancy using scatter plots.
7. Interpret the results and document analytical limitations.

Tools used

- Python
- pandas
- Matplotlib
- Seaborn
- Jupyter Notebook
- Visual Studio Code

FINDINGS

1. Life expectancy trends across countries



The line chart compares life expectancy at birth across the six countries over the study period.

Interpretation

The chart shows that life expectancy increased across all six countries between 2000 and 2015, though the pace and pattern of improvement vary by country.

Observed findings:

Country with the highest life expectancy at the end of the period: Germany

Country with the lowest life expectancy at the end of the period: Zimbabwe

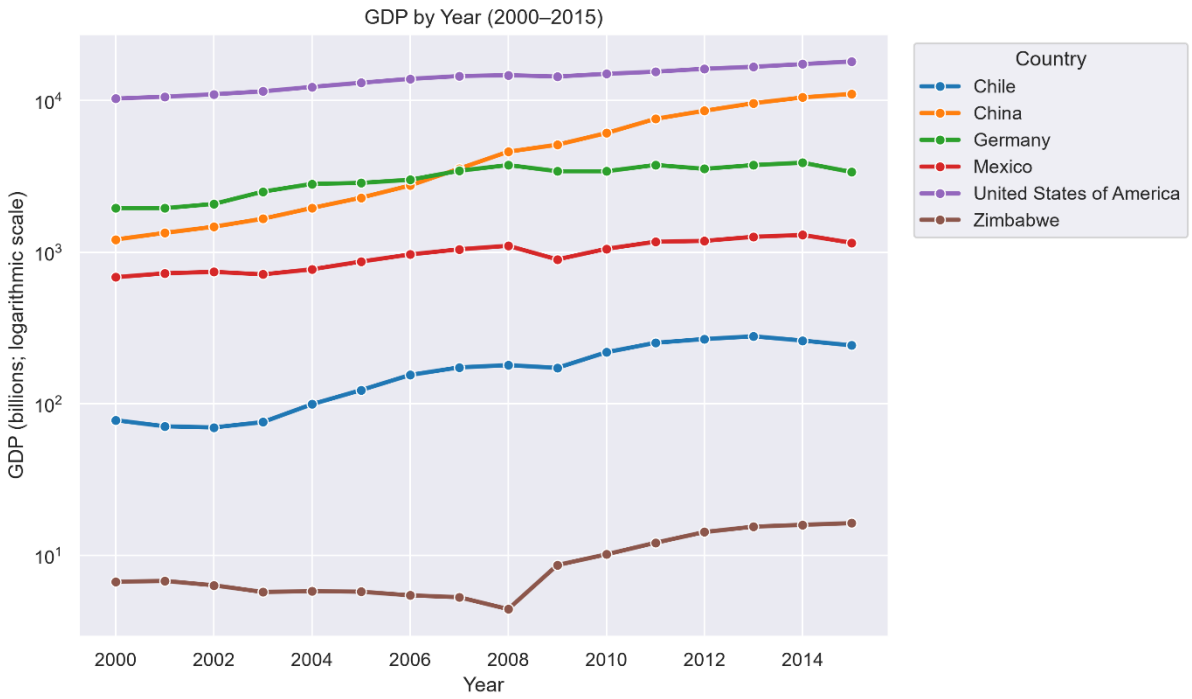
Largest absolute improvement between 2000 and 2015: Zimbabwe

Any notable periods of stagnation, decline, or rapid improvement: Zimbabwe displays a distinct pattern, with a slight decline in the early 2000s followed by a sustained and much steeper increase in later years.

Why does this matters?

Comparing life expectancy trends helps distinguish countries with relatively high life expectancy from those experiencing substantial improvements over time. A country's final level and its rate of improvement provide different perspectives on population health.

2. GDP trends over time



The GDP line chart examines changes in economic output over the study period. Because the countries have substantially different GDP levels, a logarithmic y-axis is used to make relative patterns more visible

Interpretation

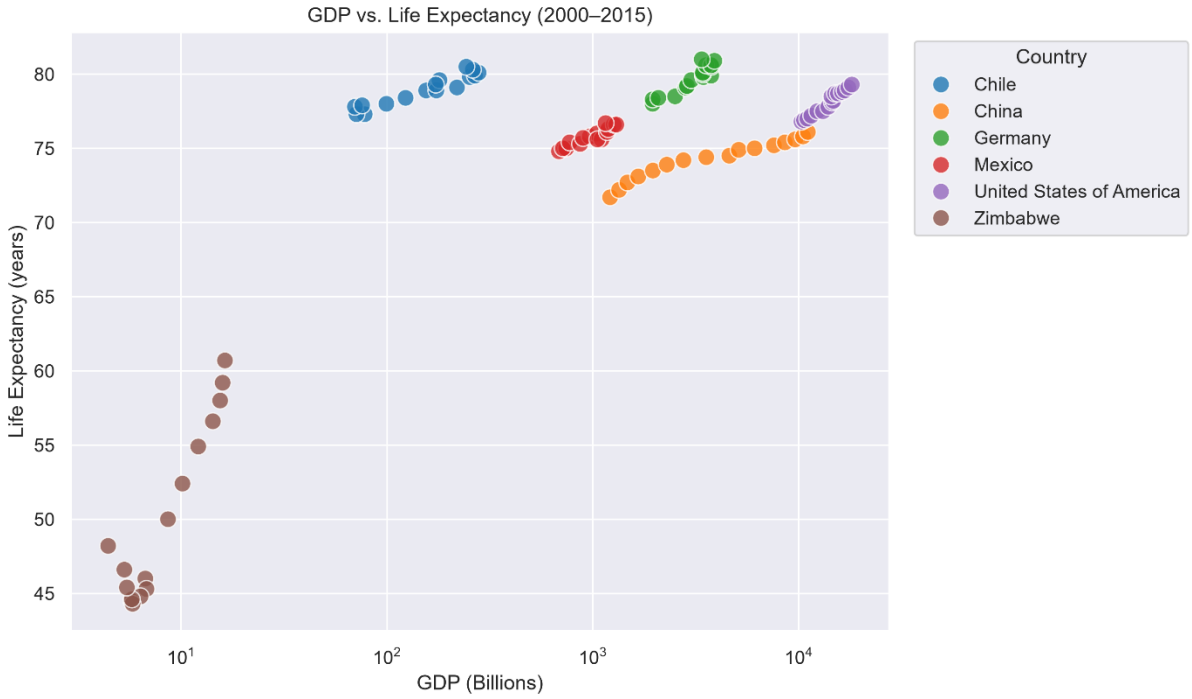
Country with the largest percentage increase in GDP: China

Country with the smallest percentage increase in GDP: Mexico

Notable differences in GDP trajectories: China, Chile, and the United States exhibited relatively smooth upward trends, while Mexico, Germany, and Zimbabwe experienced more noticeable fluctuations during the period.

Percentage growth should be distinguished from absolute GDP change. A country with a smaller initial GDP can experience a large percentage increase without having the largest increase in total economic output.

3. GDP and life expectancy



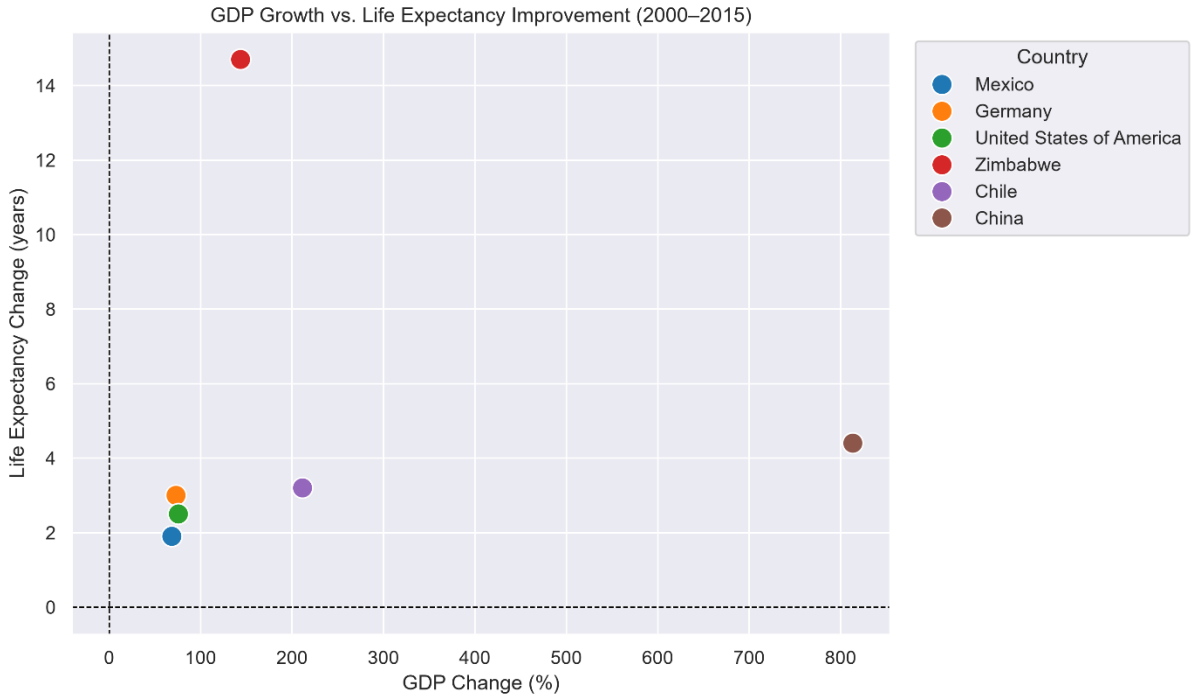
This scatter plot examines the association between GDP and life expectancy across the country-year observations.

Interpretation

This scatter plot suggests a positive association between GDP and life expectancy across all six countries. As GDP increases, life expectancy generally increases as well. However, the relationship varies by country and is not perfectly proportional.

Economic growth appears to be associated with improvements in life expectancy, but the comparison between countries such as Germany and the United States, or China and Zimbabwe, demonstrates that higher GDP is not the sole determinant of population health.

4. GDP growth versus life expectancy improvement



This country-level scatter plot contains six observations, one for each country.

Horizontal axis: Percentage change in GDP from 2000 to 2015.

Vertical axis: Change in life expectancy, measured in years.

Interpretation

This comparison examines whether countries with larger percentage increases in GDP also recorded larger improvements in life expectancy.

Countries with larger GDP increases did not necessarily experience the largest improvements in life expectancy. For example, China recorded the highest GDP growth, while Zimbabwe experienced the largest gain in life expectancy.

However, the strength of this relationship varies by country, and the chart does not establish causation.

CONCLUSIONS AND IMPLICATIONS

Main conclusions

This analysis uses a structured Python workflow to examine two development indicators across six countries over 16 years.

The analysis highlights patterns in economic growth and life expectancy across six countries between 2000 and 2015:

1. **Life expectancy and GDP provide complementary perspectives**

Life expectancy describes a health-related outcome, whereas GDP measures economic output. Examining both indicators helps reveal differences that may be overlooked when considering either measure in isolation.

2. **Levels and changes tell different stories**

A country with high life expectancy may experience a relatively small improvement, while another country starting from a lower level may experience a larger increase. Similarly, absolute GDP growth and percentage GDP growth can produce different country rankings.

3. **Relationships require careful interpretation**

A visual association between GDP and life expectancy is a starting point for further investigation. It does not establish a causal mechanism or show that GDP alone explains changes in population health.

Potential applications

The analytical workflow used here can be adapted to other domains where multiple indicators must be compared over time, including:

- Energy consumption and economic activity.
- Manufacturing output and quality indicators.
- Equipment condition monitoring and operational performance.
- Production efficiency and resource utilization.

These are potential applications of the methodology, not conclusions drawn from this dataset.

Recommended next steps

To strengthen the analysis, future work could:

- Add GDP per capita to account for population size when comparing economic conditions.
- Verify GDP units, price basis, and purchasing-power comparability from the source documentation.
- Include additional countries and a longer period to assess whether the observed patterns persist.
- Investigate potential confounding variables, such as healthcare expenditure, education, income inequality, and demographic structure.
- Apply appropriate statistical methods to quantify uncertainty and account for repeated country-level observations.

LIMITATIONS AND FINAL SUMMARY

Limitations

Limited country coverage - The dataset contains six countries. The observed patterns cannot be assumed to represent all countries or the global population.

Limited time period - The analysis covers 2000–2015. It does not capture developments after 2015.

GDP measurement - Total GDP does not measure average income, individual welfare, or the distribution of economic resources. Comparisons also depend on the units, price basis, and currency treatment used in the source data.

Association does not establish causation - GDP and life expectancy may move together because of other factors. This analysis does not isolate the effects of economic growth from healthcare, education, policy, demographics, or other influences.

Repeated observations - The GDP-versus-life-expectancy scatter plot uses country-year observations that are not necessarily independent. Long-term trends and persistent differences between countries can affect correlation estimates.

Exploratory scope - The project focuses on inspection, descriptive comparisons, and visualization. It is not a causal study and does not establish a predictive model.

Final summary

This project demonstrates a reproducible approach to inspecting, preparing, exploring, and visualizing data with Python.

By comparing GDP and life expectancy across six countries, the analysis illustrates how charts and descriptive comparisons can identify patterns, reveal differences, and generate questions for further investigation.

The key contribution is not a claim that economic growth causes improvements in health. It is a transparent analytical workflow that turns a dataset into interpretable evidence while recognizing the limitations of the available data.

Project deliverables

- Python notebooks documenting the analysis.
- A cleaned dataset.
- Reproducible visualizations.
- A GitHub README describing the project.

This case study report.

- Repository:
  - https://github.com/TekMasterSciences/life-expectancy-gdp-analysis.git
- Contact:
  - Email. adityavg86@gmail.com
  - LinkedIn: www.linkedin.com/in/aditya-verma-gottumukkala